
Sequential Product Recommendation in E-Commerce with Implicit Feedback and Category-Based Hybrid Models

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Abstract

Sequential recommender systems in e-commerce face a critical challenge: the majority of users have very short interaction histories, limiting the effectiveness of purely behavioral models. In this work, we study next-item recommendation on the RetailRocket dataset, where 85.79% of users have two or fewer interactions, and propose the **Adaptive Hierarchical Hybrid (AHH)** model. AHH combines four complementary signals—sequential transitions, category popularity, hierarchical category backoff, and global popularity—whose contributions are dynamically weighted according to the confidence of the user’s most recent sequential signal and the length of their interaction history. We evaluate AHH against seven baselines, including ItemKNN and GRU4Rec, using Recall@10, nDCG@10, Novelty@10, Category Diversity@10, and Catalog Coverage@10 on a strictly held-out test set. AHH achieves Recall@10=0.1144 globally and 0.1263 for short-history users, outperforming the strongest sequential baseline by +11.57% and GRU4Rec by +596%. These results suggest that in high-sparsity implicit feedback settings, category metadata and hierarchical backoff are more informative than neural sequence modeling capacity. Code available at: <https://github.com/MaxiNavia/AHHsysrec26>

1. Introduction

E-commerce recommender systems must operate under conditions that challenge standard collaborative filtering and sequence modeling approaches: users interact with products through implicit signals (views, add-to-cart, purchases)

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rather than explicit ratings, and a large fraction of users have extremely short histories. When fewer than two interactions are available per user, purely sequential or collaborative models have insufficient signal to produce reliable recommendations.

This paper addresses next-item recommendation in the RetailRocket dataset (RetailRocket, 2016), an e-commerce dataset characterized by high sparsity and rich product metadata including a category tree. Our central question is: *Can category metadata improve sequential recommendation for users with short interaction histories?*

We propose the Adaptive Hierarchical Hybrid (AHH) model, which explicitly leverages the category tree to fall back gracefully from fine-grained sequential signals to coarser category-level signals when user history is sparse. The key novelty is the adaptive weighting function that adjusts the contribution of each signal based on the confidence of the most recently observed sequential transition and the length of the user’s history, without requiring any explicit user features or additional training beyond the interaction log.

Our contributions are:

- A hybrid scoring function combining sequential transitions, direct category popularity, hierarchical category backoff, and global popularity with per-user adaptive weights.
- An empirical evaluation showing that AHH outperforms all baselines including GRU4Rec on a high-sparsity e-commerce dataset.
- A finding that neural sequence models underperform simple category-based hybrids in extreme sparsity settings.
- Analysis of the relevance-diversity tradeoff across all evaluated models.

2. Related Work

2.1. Sequential Recommendation

Session-based recommendation with recurrent neural networks was pioneered by Hidasi et al. (2015), who introduced

GRU4Rec, a model that processes sequences of item interactions using Gated Recurrent Units. Subsequent work introduced attention mechanisms (SASRec (Kang & McAuley, 2018)), achieving state-of-the-art performance on datasets with sufficiently long interaction sequences. However, these models require multiple interactions per user to build meaningful representations, a condition not satisfied in highly sparse datasets.

2.2. Implicit Feedback

Hu et al. (2008) formalized collaborative filtering for implicit feedback, proposing Weighted Alternating Least Squares (WALS) where different interaction types (view, click, purchase) are assigned confidence weights. We follow this weighting scheme in our preprocessing.

2.3. Category-Aware Recommendation

Category metadata has been used to address cold-start and sparsity in several works. Hierarchical category structures have been exploited to navigate from specific to general item groups when fine-grained signals are unavailable. Our work differs in that we use the hierarchy as a dynamic backoff mechanism integrated into a hybrid scoring function, rather than as a standalone recommendation signal.

2.4. Hybrid Models

Hybrid approaches combining collaborative and content-based signals have a long history in recommender systems (Ricci et al., 2010). Our contribution is specifically in the adaptive weighting mechanism that dynamically shifts the balance between signals based on per-user sequence confidence, without requiring a separate learning phase.

3. Dataset

We use the **RetailRocket** dataset (RetailRocket, 2016), which contains real user interactions from an e-commerce platform recorded over 4.5 months. The dataset includes three event types—*view*, *addtocart*, and *transaction*—along with item property metadata and a category tree.

The extreme sparsity of this dataset—where more than 85% of users have at most two interactions—motivates our focus on short-history users and the use of category metadata as a complementary signal.

Since explicit ratings are absent, we treat all interactions as implicit feedback and assign confidence weights based on interaction type: $w(\text{view}) = 1$, $w(\text{addtocart}) = 3$, $w(\text{transaction}) = 5$. This follows the weighting scheme of Hu et al. (2008).

4. Methodology

4.1. Problem Formulation

We formulate the task as **next-item recommendation**. Given a user u with interaction history $H_u = \langle i_1, i_2, \dots, i_{|H_u|} \rangle$ ordered by timestamp, the goal is to produce a ranked list of K items most likely to be the next item $i_{|H_u|+1}$ that u will interact with.

4.2. Evaluation Protocol

We adopt **temporal leave-one-out** splitting: for each user, the last interaction is reserved for test and all preceding interactions form the training set. To prevent data leakage during hyperparameter tuning, the 50,000 sampled evaluation users are split 50/50 into a validation set (used exclusively for tuning) and a held-out test set (used exclusively for final metric reporting). No model sees test users during parameter selection.

We evaluate at $K = 10$ using: **Recall@K** (fraction of users for whom the target item appears in the top- K), **nDCG@K** (normalized Discounted Cumulative Gain), **Novelty@K** (average $-\log_2(p_i)$, higher = more novel), **Category Diversity@K** (proportion of distinct categories in the top- K), and **Catalog Coverage@K** (fraction of the item catalog that appears in at least one user’s top- K list).

4.3. Baseline Models

We compare against: **Random**; **Most Popular** (globally top items by weighted interaction count); **Category Popularity** (top items within the user’s most recent category); **ItemKNN** (implicit item-item co-occurrence with cosine similarity and recency decay $1/\text{rank}$); **Sequential Transition** (first-order Markov model on the last observed item); **GRU4Rec** (Hidasi et al., 2015) (GRU-based RNN, top-50k vocabulary, embedding 128, hidden 256, 10 epochs, $\text{lr } 10^{-3}$); and **Fixed Hybrid Seq+Category** ($\alpha = 0.7$ Sequential + $\beta = 0.3$ Category Popularity, ablation for adaptive weighting).

Table 1. RetailRocket dataset statistics.

Attribute	Value
Total interactions	2.76M
Unique users	1.41M
Event types	view, addtocart, transaction
Time span	4.5 months
Users with 1 interaction	71.15%
Users with ≤ 2 interactions	85.79%
Items with known category	90.73%

4.4. Proposed Model: Adaptive Hierarchical Hybrid

The AHH model combines four signals into a unified scoring function:

$$\text{score}(u, i) = w_{\text{seq}} \cdot s_{\text{seq}}(u, i) + w_{\text{cat}} \cdot s_{\text{cat}}(u, i) + w_{\text{hier}} \cdot s_{\text{hier}}(u, i) + w_{\text{pop}} \cdot s_{\text{pop}}(i) \quad (1)$$

Sequential score $s_{\text{seq}}(u, i)$. Candidates are ranked by their weighted co-occurrence count after i_{last} in training sequences, i.e. $C[i_{\text{last}}][i] = \sum_{u'} \sum_t w_{t+1} \cdot \mathbf{1}[i_t^{u'} = i_{\text{last}}, i_{t+1}^{u'} = i]$. The ranked list is scored via reciprocal-rank weighting:

$$s_{\text{seq}}(u, i) = \frac{w_{\text{seq}}}{\text{rank}_{\text{seq}}(i)} \quad (2)$$

When no transitions exist for i_{last} , the model falls back to the global popularity ranking.

Category score $s_{\text{cat}}(u, i)$. Scores candidate i by its implicit feedback-weighted popularity within the most recently observed category $c_{\text{last}}(u)$:

$$s_{\text{cat}}(u, i) = \frac{\sum_{e \in \text{events}(i, c_{\text{last}})} w(e)}{\max_{j \in c_{\text{last}}} \sum_{e \in \text{events}(j, c_{\text{last}})} w(e)} \quad (3)$$

This score is zero for items outside $c_{\text{last}}(u)$.

Hierarchical score $s_{\text{hier}}(u, i)$. AHH traverses the category tree upward from $c_{\text{last}}(u)$, collecting the popularity-ranked item lists at each ancestor level (from most specific to most general). The concatenated candidate list is scored via reciprocal-rank weighting:

$$s_{\text{hier}}(u, i) = \frac{w_{\text{hier}}}{\text{rank}_{\text{hier}}(i)} \quad (4)$$

Items from more distant ancestors appear later in the list, receiving implicitly lower scores without requiring an explicit depth penalty.

Global popularity score $s_{\text{pop}}(i)$. Normalized global implicit feedback-weighted popularity across all users:

$$s_{\text{pop}}(i) = \frac{\sum_u \sum_{e \in \text{events}(u, i)} w(e)}{\max_j \sum_u \sum_{e \in \text{events}(u, j)} w(e)} \quad (5)$$

This serves as a final fallback when all other signals are weak.

Adaptive weight function. The weights $(w_{\text{seq}}, w_{\text{cat}}, w_{\text{hier}}, w_{\text{pop}})$ are computed per user from the sequential confidence $c_{\text{sep}}(u)$ and history length $|H_u|$.

$c_{\text{sep}}(u)$ measures how reliably i_{last} predicts the next item—computed as the log-scaled total outgoing transition weight of i_{last} , normalized by s_{scale} (90th percentile of item transition strengths):

$$c_{\text{sep}}(u) = \min\left(1, \frac{\log(1 + \text{str}(i_{\text{last}}))}{\log(1 + s_{\text{scale}})}\right) \quad (6)$$

where $\text{str}(i_{\text{last}}) = \sum_j C[i_{\text{last}}][j]$ is the total weighted outgoing transitions from i_{last} .

This is blended with a history-length confidence

$$c_{\text{hist}}(u) = \min\left(1, \frac{\log(1 + |H_u|)}{\log(1 + 10)}\right) \quad (7)$$

into a composite confidence:

$$\sigma(u) = \alpha \cdot c_{\text{sep}}(u) + (1 - \alpha) \cdot c_{\text{hist}}(u), \quad \alpha = \text{transition_mix} = 0.75 \quad (8)$$

The adaptive weights are then:

$$w_{\text{seq}}(u) = \text{seq_min} + \text{seq_span} \cdot \sigma(u) \quad (9)$$

$$w_{\text{cat}}(u) = \text{cat_base} - \text{cat_drop} \cdot \sigma(u) \quad (10)$$

$$w_{\text{hier}}(u) = \text{hier_max} \cdot (1 - c_{\text{sep}}(u))^\gamma \quad (11)$$

$$w_{\text{pop}}(u) = \text{pop} \quad (\text{constant}) \quad (12)$$

Note that w_{hier} uses $c_{\text{sep}}(u)$ directly rather than $\sigma(u)$, since hierarchical backoff is most needed when the last item specifically has unreliable transitions, regardless of overall history length. All weights are clipped to $[0, \infty)$ and normalized to sum to 1. The full tuple $(w_{\text{seq}}, w_{\text{cat}}, w_{\text{hier}}, w_{\text{pop}}) = f(\sigma(u), c_{\text{sep}}(u))$ is determined entirely by observable properties of the training history, requiring no additional learning.

4.5. Hyperparameter Tuning

We tune six hyperparameters on the validation set via exhaustive grid search (Table 2). Selection maximizes Recall@10 on 25,000 validation users; final metrics are reported on 25,000 disjoint test users.

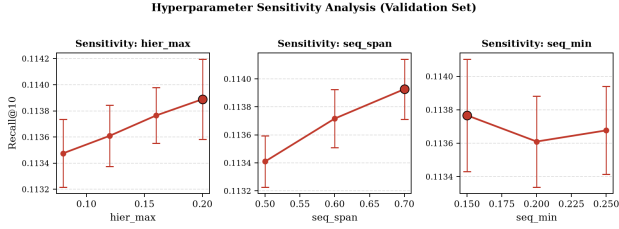
5. Parameter Analysis

Figure 1 shows the sensitivity of Recall@10 on the validation set to the three most impactful parameters: `hier_max`, `seq_min`, and `seq_span`.

The most sensitive parameter is `hier_max`: increasing the maximum weight for the hierarchical signal monotonically improves Recall@10 up to 0.20, confirming that the category tree provides useful signal beyond direct category

Table 2. Hyperparameter grid and selected values.

Parameter	Grid	Selected
seq_min	{0.15, 0.20, 0.25}	0.15
seq_span	{0.50, 0.60, 0.70}	0.70
cat_base	{0.45, 0.55, 0.65}	0.55
cat_drop	{0.20, 0.30, 0.40}	0.30
hier_max	{0.08, 0.12, 0.16, 0.20}	0.20
transition_mix	{0.50, 0.75, 1.00}	0.75


 Figure 1. Hyperparameter sensitivity on the validation set. Each panel shows mean Recall@10 $\pm$ std across all other parameter combinations.

popularity. `seq_span` shows a similar monotonic trend. `seq_min` is less sensitive (best value 0.15), suggesting that even users with weak sequential confidence benefit from a small baseline sequential weight.

6. Results

6.1. Global Performance

Table 3 reports all metrics for the full test set. AHH achieves the highest Recall@10 (0.1144) and nDCG@10 (0.0686), outperforming the next best model (Fixed Hybrid Seq+Cat) by +4.4% in Recall@10 and the strongest single-signal baseline (Sequential Transition) by +11.57%. The improvement over ItemKNN is +44.67% and over Category Popularity +52.53%.

6.2. Short-History Users

Table 4 reports results for users with at most two interactions ($|H_u| \leq 2$), representing 85.79% of the user base and the primary motivation for this work. AHH achieves Recall@10=0.1263, a gain of +12.77% over Sequential Transition (+0.0143 absolute) and +45.51% over ItemKNN. The relative improvement over the sequential baseline is larger in the short-history regime than globally, confirming that the hierarchical category signal is most valuable precisely when interaction history is sparse.

6.3. Ablation Study

Figure 2 shows Recall@10 for each component combination. Category Popularity alone (0.0750) does not surpass Sequential Transition (0.1025), confirming that the sequential signal is the stronger single signal. Combining both (Fixed Hybrid) yields 0.1096 (+6.9% over sequential alone), and adding the hierarchical backoff with adaptive weighting achieves the best result (0.1144). Each component contributes incrementally, with the hierarchical signal providing the final +4.4% gain over the fixed hybrid.

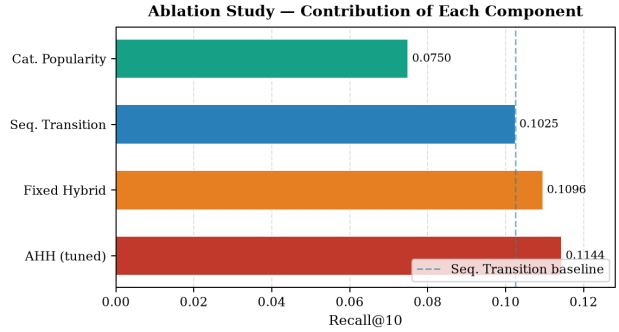


Figure 2. Ablation study: Recall@10 per component combination.

6.4. GRU4Rec Analysis

GRU4Rec achieves Recall@10=0.0164 globally and 0.0173 for short-history users, placing it below all non-trivial baselines. The training loss decreased from 10.23 at epoch 1 to 8.64 at epoch 30, with no further improvement beyond epoch 20, indicating convergence. We attribute this to the dataset’s extreme sparsity: with 85.79% of users having at most two interactions, GRU4Rec cannot generalize beyond memorizing local item-item transitions, a task in which the simpler Sequential Transition model is already competitive. This is consistent with prior observations that neural sequential models require sufficiently long sequences to outperform Markov-based methods. GRU4Rec achieves the highest Category Diversity@10 (0.931), reflecting its vocabulary-based scoring rather than accurate personalization.

We also attempted SASRec (Kang & McAuley, 2018) as an additional baseline; it failed to learn meaningful representations under extreme sparsity (Recall@10=0.0014), confirming that attention-based sequential models require denser interaction histories. A thorough evaluation of SASRec with pre-training or data augmentation is left for future work.

Table 3. Global results on the held-out test set (25,000 users). Bold values indicate the best result per column.

Model	Recall@10	nDCG@10	Novelty@10	Cat.Div@10	Coverage@10
AHH (tuned)	0.1144	0.0686	13.60	0.507	0.230
Fixed Hybrid Seq+Cat	0.1096	0.0685	13.16	0.623	0.228
Sequential Transition	0.1025	0.0664	13.24	0.730	0.256
ItemKNN	0.0791	0.0443	15.42	0.741	0.371
Category Popularity	0.0750	0.0399	13.51	0.070	0.059
GRU4Rec	0.0164	0.0100	13.78	0.931	0.143
Most Popular	0.0022	0.0009	10.20	0.955	0.000
Random	0.0000	0.0000	18.78	0.995	0.810

 Table 4. Results for short-history users ($|H_u| \leq 2$). Bold values indicate the best result per column.

Model	Recall@10	nDCG@10	Novelty@10	Cat.Div@10
AHH (tuned)	0.1263	0.0763	13.53	0.503
Fixed Hybrid Seq+Cat	0.1209	0.0763	13.05	0.640
Sequential Transition	0.1120	0.0737	13.10	0.744
ItemKNN	0.0868	0.0498	15.08	0.755
Category Popularity	0.0832	0.0447	13.46	0.090
GRU4Rec	0.0173	0.0107	13.58	0.938
Most Popular	0.0025	0.0011	10.20	0.956
Random	0.0000	0.0000	18.79	0.995

6.5. Relevance–Diversity Tradeoff

Figure 3 shows the relationship between Recall@10 and Category Diversity@10 across all models. There is a clear negative correlation: models that prioritize category-specific popularity (AHH, Category Popularity) achieve higher recall but lower diversity, while models that spread recommendations across the catalog (Random, Most Popular, GRU4Rec) achieve higher diversity at the cost of recall. ItemKNN occupies an intermediate position (Recall@10=0.0791, Cat.Div@10=0.741). AHH achieves the best recall at the cost of moderate diversity (Cat.Div@10=0.507), an appropriate tradeoff for an e-commerce next-item prediction objective.

6.6. Qualitative Analysis

Consider user 474893 with $|H_u| = 2$, whose two training interactions are items in category 1261, and whose test item (401151) also belongs to category 1261. The sequential confidence c_{sep} is low because item 191555 (last observed) has few reliable outgoing transitions. The resulting adaptive weights are approximately: $w_{\text{seq}} \approx 0.33$, $w_{\text{cat}} \approx 0.46$, $w_{\text{hier}} \approx 0.16$, $w_{\text{pop}} \approx 0.05$.

- **Sequential Transition:** misses item 401151—the transition from 191555 to 401151 is not observed frequently in training.
- **GRU4Rec:** also misses. With only two interactions, the GRU cannot build a meaningful user representa-

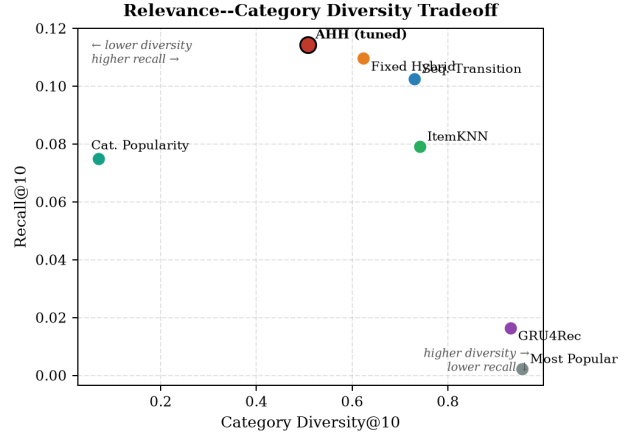


Figure 3. Relevance–diversity tradeoff. Models with higher category diversity tend to have lower recall, revealing a tension between personalization accuracy and intra-list diversity.

tion.

- **Category Popularity:** hits. Item 401151 is popular within category 1261.
- **AHH (tuned):** hits. By assigning high weight to the category signal ($w_{\text{cat}} \approx 0.46$) and incorporating hierarchical backoff, AHH recovers item 401151 even though the sequential signal is unreliable.

7. Conclusions

We presented the Adaptive Hierarchical Hybrid (AHH) model for next-item recommendation in a high-sparsity implicit feedback e-commerce dataset. Our main findings are:

1. Category metadata significantly complements sequential signals for short-history users. AHH outperforms Sequential Transition by +11.57% globally and +12.77% for $|H_u| \leq 2$ users.
2. The hierarchical category backoff provides a consistent additional gain (+4.4%) over a fixed-weight hybrid, confirming that the category tree encodes useful proximity information.
3. GRU4Rec underperforms all non-trivial baselines (Recall@10=0.0164), confirming that neural sequence models require sufficient interaction history to generalize.
4. There is a clear relevance-diversity tradeoff: AHH achieves the highest recall at the cost of moderate category diversity, consistent with the e-commerce next-item prediction objective.

Future work could explore learning the weighting function from data (e.g., via a small meta-learner), incorporating temporal decay into the sequential and category signals, and applying LLM-based semantic embeddings for category similarity to generalize across related but non-identical categories.

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Impact Statement

This paper presents work on recommendation systems for e-commerce. The primary application is improving product discovery for users with limited interaction history. We do not foresee specific ethical concerns beyond standard considerations for personalization systems (filter bubbles, popularity bias), which are partially addressed by our diversity analysis.

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